

Project Title

Automated Network Request Management in ServiceNow

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Project overview:

The **Automated Network Request Management in ServiceNow** project focuses on simplifying and accelerating the handling of network-related service requests. By leveraging ServiceNow's service catalog, workflows, and automation capabilities, the solution eliminates manual bottlenecks and ensures requests are processed efficiently. End users can easily submit requests through a self-service portal, while automated routing, approvals, and task assignments streamline fulfillment. Where possible, integration with network automation tools enables end-to-end execution without human intervention. This results in faster turnaround, improved SLA compliance, and enhanced transparency for both users and IT teams.

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1) Introduction:

Modern enterprises rely heavily on robust and efficient network services to support day-to-day business operations. As organizations grow, the demand for network-related requests—such as access provisioning, configuration changes, and connectivity support—also increases. Traditional manual request handling often leads to delays, errors, and limited visibility, impacting both productivity and user satisfaction.

To address these challenges, the **Automated Network Request Management in ServiceNow** project introduces a streamlined solution that leverages ServiceNow's workflow engine, service catalog, and automation features. By enabling end users to submit requests through a self-service portal and automating approvals, task assignments, and notifications, the system ensures faster, more accurate, and transparent request management.

This initiative not only enhances the user experience but also empowers IT teams by reducing repetitive tasks, improving SLA compliance, and allowing focus on strategic, high-value activities.

2) Project Objective:

This project aims to design and implement a streamlined, automated solution for managing network related service requests within ServiceNow. It enables end users to submit requests for network services through a user-friendly self-service portal.

The system leverages ServiceNow's workflow engine, catalog items, and approval processes to ensure requests are properly captured, validated, and routed for fulfilment. Upon submission, requests trigger automated notifications, task assignments, and—where applicable—integration with network automation tools or scripts to fulfil standard requests without manual intervention.

3) Key Features:

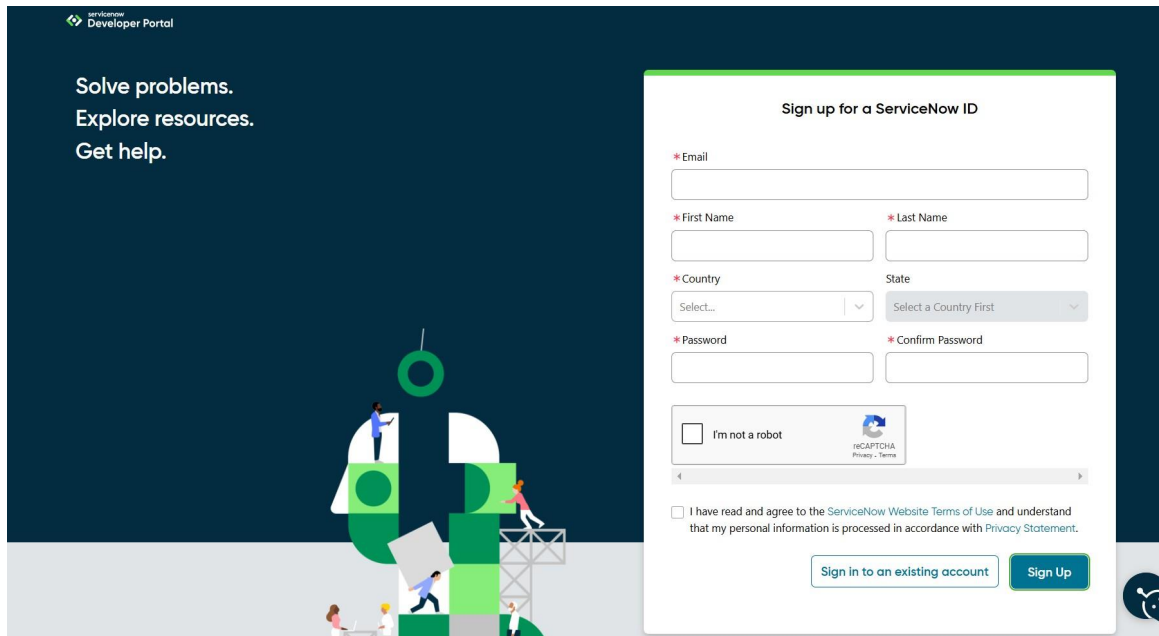
- Custom service catalog for common network requests
- Dynamic forms to capture relevant request details
- Automated approval workflows based on request type and sensitivity
- Integration with infrastructure management or orchestration tools (optional)
- Real-time status updates and notifications to requesters and technicians

Reporting and analytics on request volume, resolution time, and SLA adherence

4) ServiceNow Developer Setup:

Create a Developer Account

1. Go to ServiceNow Developer Portal(<https://developer.servicenow.com/dev.do>).
Sign up for a free developer account and fill out the following details.



ServiceNow Developer Portal

Solve problems.
Explore resources.
Get help.

Sign up for a ServiceNow ID

* Email

* First Name * Last Name

* Country State

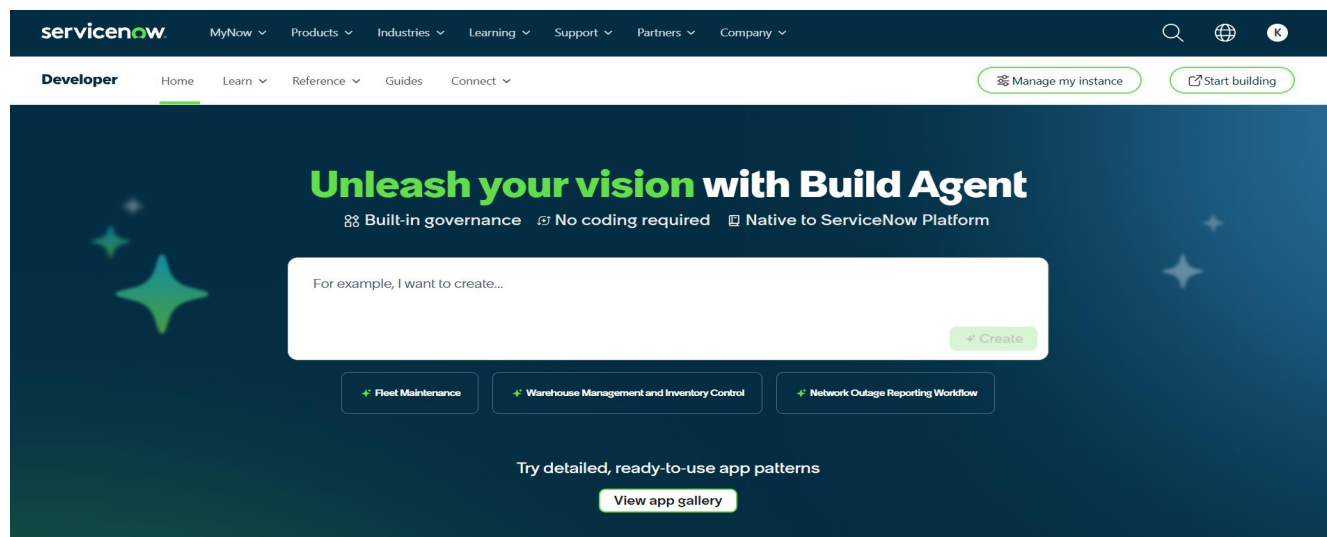
* Password * Confirm Password

☐ I'm not a robot 

☐ I have read and agree to the [ServiceNow Website Terms of Use](#) and understand that my personal information is processed in accordance with [Privacy Statement](#).

[Sign in to an existing account](#) [Sign Up](#)

2. After signing up, you will get a verification mail to your provided email id. After the verification your ServiceNow Developer Portal Home Page will appear.
3. Now click on start building it will take you to the section where you can request a Personal Developer Instance (PDI) or start using App Engine Studio and other tools.
4. Profile Icon (Top Right Corner) → Manage your account, request instances, and check your developer profile.



servicenow MyNow ▾ Products ▾ Industries ▾ Learning ▾ Support ▾ Partners ▾ Company ▾

Developer Home Learn ▾ Reference ▾ Guides Connect ▾ [Manage my instance](#) [Start building](#)

Unleash your vision with Build Agent

⚙️ Built-in governance ⚙️ No coding required ⚙️ Native to ServiceNow Platform

For example, I want to create...

[+ Create](#)

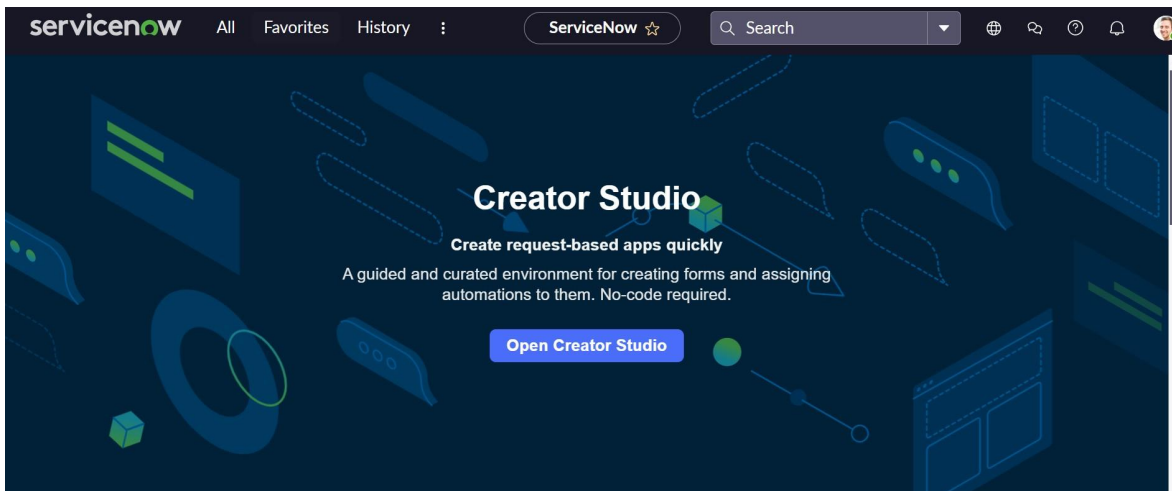
[+ Fleet Maintenance](#) [+ Warehouse Management and Inventory Control](#) [+ Network Outage Reporting Workflow](#)

Try detailed, ready-to-use app patterns

[View app gallery](#)

5) Project Implementation in ServiceNow:

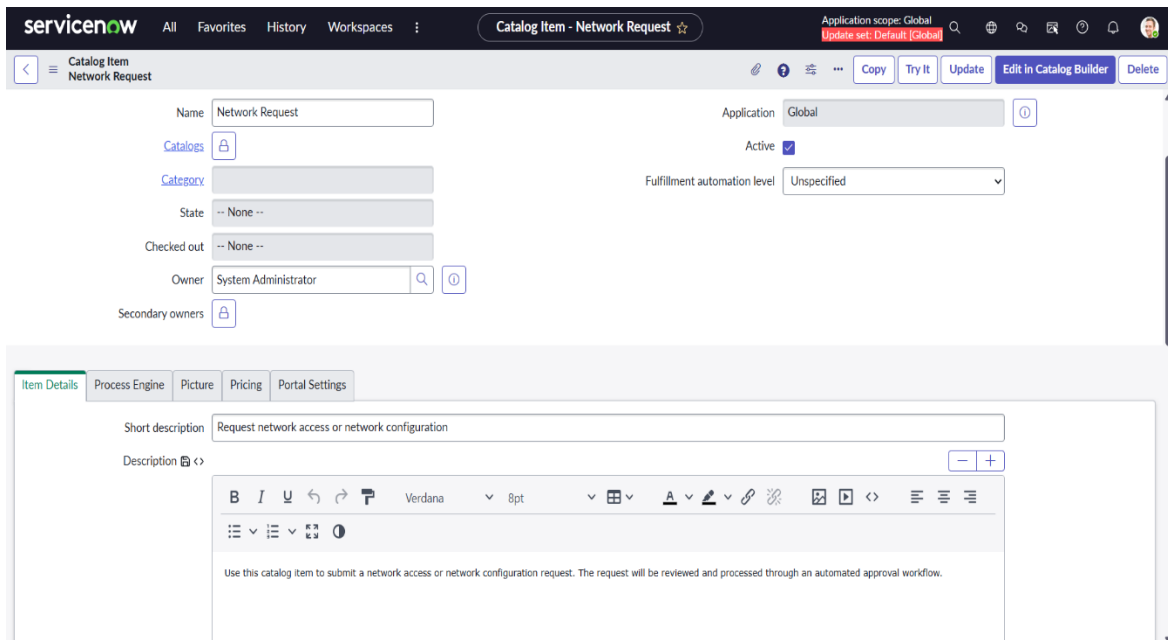
After the instance building is completed, the page will be directed to your creator studio. Creator Studio in ServiceNow provides a guided, no-code environment to build applications quickly. It is especially useful for creating request-based applications by defining forms, setting up tables, and automating workflows.



a. Service Catalog Creation:

i. Creation of Service Catalog

1. Navigate to Application navigator
2. Click on All → search for Service Catalog
3. Under Service Catalog → Maintain items
4. Click on New
5. Fill the details → Name– Network Request
6. Select Catalog → Service Catalog
7. Select Category → Network
8. Fill the Short Description as Network request Management
9. Click on Save.



servicenow All Favorites History Workspaces Catalog Item - Network Request Application scope: Global Update set: Default (Global)

Catalog Item Network Request Copy Try It Update Edit in Catalog Builder Delete

Name Network Request Application Global

Catalogs Catalog Category State -- None -- Active ☒ Fulfillment automation level Unspecified

Checked out -- None -- Owner System Administrator

Secondary owners

Item Details Process Engine Picture Pricing Portal Settings

Short description Request network access or network configuration

Description

B I U ↶ ↷ Verdana 8pt

Use this catalog item to submit a network access or network configuration request. The request will be reviewed and processed through an automated approval workflow.

ii. Variables Configuration

Open the catalog item just created.

Scroll down to the Variables related list and click New to create form fields.

1. Select Variables type as Single, Multi line text, reference, choices etc as per requirement
2. Catalog item – Network Request
3. Order–100, 200, 300, ...
4. Question– provide the variable label
5. Name–provide the variables name (used for scripting)
6. Tooltip– this will appear when cursor overed on the field
7. Example text – this will suggest what we need to enter on the field.
8. Mandatory, read-only – need to configure on demand.
9. Auto populate– need to select dependent variable, apply dot walking to get selected value.
10. Click on Save or Submit.

servicenow All Favorites History Workspaces Catalog Item - Network Request Application scope: Global Update set: Default [Global]

Catalog Item Network Request [SN Utils] Versions (31)

Variables (10) Variable Sets (1) Catalog UI Policies (2) Catalog Client Scripts Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles Related Catalog Items

Assigned Topics

Order Search Actions on selected rows... New

Catalog item = Network Request

Type	Question	Order
Container Start	Service Details	100
Multiple Choice	Is this a New connection or Relocation?	200
Single Line Text	If this is a relocation , Please provide...	300
Single Line Text	If this is a relocation, Please provide...	400
Container Start	Location & Devices Type	500
Single Line Text	Please provide address here	600
Multiple Choice	Types of Devices	700
Single Line Text	Provide device details here	800
Container Start	Additional information	900
Single Line Text	If anything else, please specify	910

1 to 10 of 10

iii. Variables Types

1. Is this a new connection or Relocation? → Choice → New/ Relocation/None
2. If this is a relocation, please provide your relocated address here → String
3. Types of devices → Choice → Laptop/Mobiles/Others
4. Please provide address here → String
5. Provide device details here → String
6. If anything, else please specify → String

iv. Variable Set Configuration

To enhance form usability:

- Navigate to the Variable Sets (optional).
- Follow the same procedure as we used for Variables Creation, for the variable set as well.

- Apply variable sets to the catalog item.

Variables (10)	Variable Sets (1)	Catalog UI Policies (1)	Catalog Client Scripts	Available For	Not Available For	Categories (1)	Catalogs (1)	Catalog Data Lookup Definitions	Related Articles	Related Catalog Items
Assigned Topics										
Order Search ⊙ 🔗 — Actions on selected rows... New Edit...										
Catalog Item = Network Request										
Variable set Order ▲										
Requester information										
1 to 1 of 1										

Variables Types

1. Opened on behalf of → Reference → Reference to user table
2. Email Id → Single line text → Auto populate by Opened on behalf of variable.
3. User name → Single line text → Auto populate by Opened on behalf of variable.
4. Phone Number → Single line text → Auto populate by Opened on behalf of variable.
5. Proof of Document → Attachment

Variables (5)

Catalog UI Policies

Catalog Client Scripts

Included In (1)

Catalog Data Lookup Definitions

≡

▽

Order

Search

⊙

🔗

—

Actions on selected rows...

New

Variable set = Requester information

☐ Q Name	Type	Question	Order ▲
email_id	Single Line Text	Email Id	
proof_of_document	Attachment	Proof of Document	
opened_on_behalf_of	Reference	Opened on behalf of	
username	Single Line Text	User name	
phonenumber	Single Line Text	Phone Number	

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1 to 5 of 5

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⏩

v. Catalog UI Policy Configuration

Scenario: If user selects types of devices is Others, then Please specify field should populate.

Procedure:

1. Navigate to catalog items
2. Open Network Request item
3. In related list, we have Catalog UI policy
4. Click on New button to configure New UI policy
5. Select Applies to as Catalog item
6. Select catalog item as Network Request

7. Provide short description, if required
8. Apply condition → types of devices is others
9. Click on save, after saving the form will get UI policy actions in the related list
10. Click on New button to configure new UI Policy action, and Select the variable which we want to display on condition
11. Make Visible True as per our requirement
12. Update the UI Policy and Test the same on Catalog form

Variables (10)	Variable Sets (1)	Catalog UI Policies (2)	Catalog Client Scripts	Available For	Not Available For	Categories (1)	Catalogs (1)	Catalog Data Lookup Definitions	Related Articles	Related Catalog Items
----------------	-------------------	-------------------------	------------------------	---------------	-------------------	----------------	--------------	---------------------------------	------------------	-----------------------

Assigned Topics

Order

Search

⊙

🔗

—

Actions on selected rows...

New

Catalog item = Network Request

☐	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	Types of devices is others	(empty)		true	true	false	2025-09-17 07:51:06	100
☐	Relocation fields hiding	(empty)		true	true	false	2025-09-18 02:07:31	100

1 to 2 of 2

b. Creation of Table:

i. Creation of Table

Navigate to: System Definition → Tables.

- a. Click New to create a new table.
- b. Fill in Table Information:
 - i. Name: Name of the table
 - ii. Label: Backend name of the table
 - iii. Auto-generate schema: Leave it checked if you'd like ServiceNow to auto-generate schema fields.
- c. Click Submit to create the table.

servicenow All Favorites History Workspaces Admin **Table - Network Database** ☆ Application scope: Global Update set: Default [Global]

< = Table **Network Database**

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Application ⓘ

* Name

Remote Table ☐

Is Data Interface ☐

Columns Controls Application Access

≡ ⚙️ **Table Columns** for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Created by	String	(empty)	40	false	false
Sys ID	Sys ID (GUID)	(empty)	32	false	false
Updated by	String	(empty)	40	false	false
Assigned to	Reference	User	32	false	false
Device Details	String	(empty)	40	false	false
Created	Date/Time	(empty)	40	false	false
Request Number	String	(empty)	40	false	false
Customer Document	String	(empty)	40	false	false

ii. Creation of Fields

Fields are created at the table level in ServiceNow.

1. Open Application Navigator and search for Tables.
2. Go to System Definition → Tables.
3. Search for and open the Network Database table.
4. Open the Columns section.
5. Click New to create a new field.
6. Enter the required field details and click Submit.

Table Columns for text Search						
Dictionary Entries						
Column label	Type	Reference	Max length	Default value	Display	
×	Work Status	String	(empty)	40	false	
×	Device Details	String	(empty)	40	false	
×	Requested For	String	(empty)	40	false	
×	Date of Enquiry	Date	(empty)	40	false	
	Sys ID	Sys ID (GUID)	(empty)	32	false	
	Created by	String	(empty)	40	false	
	Created	Date/Time	(empty)	40	false	
	Updated by	String	(empty)	40	false	
	Updates	Integer	(empty)	40	false	
	Updated	Date/Time	(empty)	40	false	
×	Customer Address	String	(empty)	40	false	
×	Request Number	String	(empty)	40	false	
×	Assignment Group	Reference	Group	32	false	
×	Customer Document	String	(empty)	40	false	
×	Assigned to	Reference	User	32	false	

c. Request Approvals Creation (Related List):

i. Creation of Related List

1. Navigate to System Definition → Relationships.
2. Click New to create a new relationship.
3. Fill in the following details:
 - Name: Approval Request
 - Applies to Table: Network Database table.
 - Queries from Table: Sysapprovals table.
 - Active: Make sure it's set to True.
4. Save the relationship.

Relationship Approval Request		Update Delete
Name	Approval Request	Application Global
Advanced ☐		Applies to table Network DataBase Table [u_network_da...
		Queries from table Approval [sysapproval_approver]

ii. Adding Related List to the Table

1. Navigate to Form Designer for the table where you want to show related records.
2. Add a Related List widget to the form.
3. Select the Related List you want to show

Query with  Turn on ECMAScript 2021 (ES12) mode 

```
1  (function refineQuery(current, parent) {  
2  
3      // Add your code here, such as current.addQuery(field, value);  
4      current.addQuery('source_table', parent.getTableName());  
5      current.addQuery('document_id', parent.sys_id);  
6  
7  })(current, parent);
```

d. Creation & Implementation of flows, Actions in Flow Designer :

i. Creation of Flow

1. Navigate to Flow designer home page
2. Click on New to create a new flow
3. Provide flow name as Network Request
4. Provide description of flow
5. Click on Build flow.

ii. Configuring Trigger

1. Click on (+) Icon to Configure the Trigger
2. Select Trigger as Application → Service catalog
3. Click on Done.

TRIGGER

 Service Catalog  

Trigger

Service Catalog

Advanced Options 

Delete

Cancel

Done

iii. Configuring Actions

Click on Actions button to configure new action

1. Get Catalog Variables

- Click on Action, search for Get Catalog Variables
- Select Get Catalog Variables
- Action Inputs → Trigger → service catalog → Requested Item
- Template catalog items → Select table → Network Request
- Select the Required Variables and Move to the selected area.
- Click on done

1  Get Catalog Variables from Network Request 

Action Properties

Action

Action Inputs

* Submitted Request [Requested Item]  

Select one or more values from the Template Catalog Items and Variable Sets, and select the required Catalog Variables to generate output data pills. You cannot choose the same Catalog Variable from multiple Template Catalog Items and Variable Sets.

* Template Catalog Items and Variable Sets [Catalog Items and Variable Sets]   

Catalog Variables

Available		Selected
No available values		opened_on_behalf_of
		please_provide_address_here
		types_of_devices
		email_id
		proof_of_document
		username
		phonenumber
		is_this_a_new_connection_or
		if_this_is_a_relocation_please
		if_this_is_a_relocation_please
		provide_device_details

Note: If removing a variable from the 'Selected' list, it will be moved to 'Available' list only if the variable is from the selected Template Catalog Items and Variable Sets. Otherwise, the variable is

2. Create Record

- Select action as Create Record
- Select table as Network Database
- Click on Add fields button to configure the fields
- Configure the Required fields as shown in the below picture
- Click on done

The screenshot shows a configuration window titled 'Create Network DataBase Table Record'. It has a tab '2' and a close button. The window is divided into two main sections: 'Action Properties' and 'Action Inputs'.

Action Properties: The 'Action' dropdown is set to 'Create Record'.

Action Inputs: This section contains a list of fields to be configured. Each field has a dropdown menu to select a value and a button to add or remove the field.

* Table	* Fields
Network DataBase Table [u_net...	Request Number
	Requested For
	Work Status
	Assignment Group
	Date of Enquiry
	Device Details
	Customer Address

At the bottom right, there are three buttons: 'Delete', 'Cancel', and 'Done'.

3. Ask for approvals

- Select action as Ask for Approval
- Select target record → Create record → network database table
- Provide Approval Reason → Waiting for approval
- Configure approval rules → Approve, reject, approve/reject
- Select approvals as Anyone approves, everyone approves etc.
- We can select approvals like static/dynamic as shown below
- Click on done

3 **Ask For Approval**

Action Properties

Action: **Ask For Approval**

Action Inputs

* Record: Trigger -> Requested Item ...

Table: Requested Item [sc_req_item]

Approval Reason: Waiting for approval

Approval Field: Approval

Journal Field: Approval history

* Rules

Approve When: Anyone approves Network X

Due Date: None

Add another OR rule set

OR AND

Delete Cancel Done

4. Flow Logic

- Select action as flow logic and Select If condition
- Apply condition → Ask for approvals state is Approved/Rejected as per requirement
- Click on done

If

Condition Label: If Request is Approved

* Condition 1: Trigger - Servic... → Appro... is Approved or and

Add another condition set(OR)

Delete Cancel Done

5. Send Email

- Select action as Send Email
- Select target record → Create record → network database table
- Table will be selected automatically
- Configure To, CC, BCC as per our requirements (select static/dynamic)

- Provide Subject & Body as shown in the below picture
- Click on done

then **Send Email**

Action Properties

Action: Send Email

Action Inputs

Target Record: 2 - ... > Network DataBase Ta...

Table: Network DataBase Table [u_net...

Include Watermark: ☒

* To: 1 - Get Catalog Va... > ... > Em...

CC: 2 - Create Record > ... > Email

BCC:

* Subject: Request had been created

Body:

Hello 2 - Create ... > ... > Requested ...

We have received your request with request number: 2 - Creat... > ... > Request Nu...

Sorry for the inconvenience and Your request will be resolved within 2 business working days.

Thanks for contacting us.

Network Team

Delete Cancel Done

6. Update Record

- Select action as Update Record
- Select record as → create record → network database
- Table will be selected automatically
- Configure the fields as per requirement, as shown in below
- Click on done

Update Network DataBase Table Record

Action Properties

Action: Update Record

Action Inputs

* Record: 2 - ... > Network DataBase Ta...

* Table: Network DataBase Table [u_net...

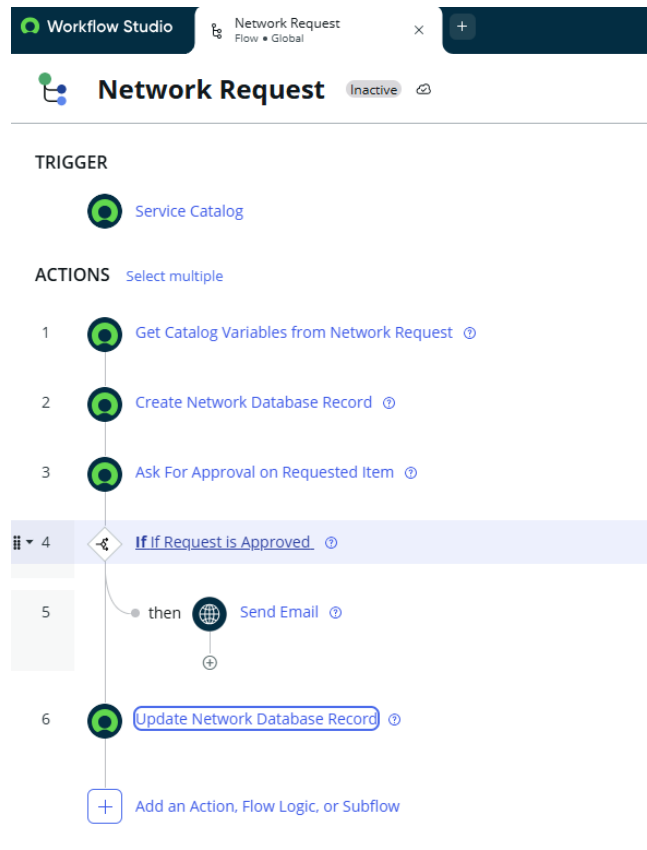
* Fields:

Assigned to	Abel Tuter
Work Status	In Progress

+ Add field value

Delete Cancel Done

iv. Flow Chart



Network Request Workflow: -

The **Network Request workflow** automates the complete request process in ServiceNow. When a user submits a request through the Service Catalog, the flow retrieves the submitted details, creates a Network Database record, and sends the request for approval. If the request is approved, an email notification is sent and the Network Database record is updated automatically.

Workflow sequence:

Service Catalog → Get Catalog Variables → Create Network Database Record → Approval → Send Email → Update Network Database Record

6. Screenshots of Output:

The screenshot shows the 'Network Request' form in the ServiceNow interface. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The breadcrumb trail is 'Service Catalog > Standard Changes > Network Standard Changes > Network Request'. The form fields include:

- Network request Management**: Email Id (text field).
- Proof of Document**: Click to add... (button).
- Opened on behalf of**: (text field with a search icon).
- User name**: (text field).
- Phone Number**: (text field).
- Is this a New connection or Relocation?**: Radio buttons for 'New Connection' (selected), 'Relocation', and 'None'.
- If this is a relocation, Please provide...**: (text field).
- Please provide address here**: (text field).
- Types of Devices**: Radio buttons for 'Laptop' (selected), 'Mobiles', and 'Others'.
- If anything else, please specify**: (text field).

On the right side, there is a sidebar with 'Order this item' (Quantity: 1, Delivery time: 2 Days), 'Order Now' (button), 'Add to Cart' (button), 'Shopping Cart' (button), and 'Emails' (button).

Fig: Network Request Service Catalog Form

The screenshot shows the 'Order Status' page in the ServiceNow interface. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The breadcrumb trail is 'Order Status: REQ0010001'. The page displays a confirmation message: 'Thank you, your request has been submitted'. Below this, the order details are shown:

- Order Placed:** 2026-08-16 23:08:03
- Request Number:** REQ0010001
- Estimated Delivery Date of Complete Order:** 2026-08-18

The table below shows the order details:

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
Network request Management	2026-08-18			1	
				Total	-

At the bottom, there are buttons for 'Back to Catalog', 'Continue Shopping', and 'Home'.

Fig: Network Request Order Status

EXECUTION DETAILS		Network Request		Test Run - Completed	Open flow	Open context record
FLOW STATISTICS		Run as: System Administrator	Open flow logs	Completed	2026-08-17 01:00:42	956ms
TRIGGER						
	Catalog Item Requested					
ACTIONS						
1	Get Catalog Variables from Network Request	Core Action	Completed	2026-08-17 01:00:42	21ms	
2	Create Record	Core Action	Completed	2026-08-17 01:00:42	7ms	
3	Ask For Approval	Core Action	Completed	2026-08-17 01:00:42	4ms	
4	If Request is Approved	Flow Logic	Evaluated - True	2026-08-17 01:00:42	912ms	
5	Send Email		Completed	2026-08-17 01:00:42	911ms	
6	Update Record	Core Action	Completed	2026-08-17 01:00:43	8ms	

Fig: Testing Flow Execution

Preview Email

Close

Hello Alejandra Prenatt

We have received your request with request number REQ0010003,

Sorry for the inconvenience and your request will be resolved with 2 business working days.

Thanks for contacting us

Network Team.

Ref:MSG0001350_0IIQLJFxcEO826lmjirx

Fig: Preview of Email Notification

7. Conclusion:

The **Automated Network Request Management in ServiceNow** project successfully provides a centralized and automated approach for managing network service requests. The solution enables users to submit requests through the Service Catalog, capture the required information, and move requests through approval and processing in a structured manner.

By using **Service Catalog and Flow Designer**, the project reduces manual effort and improves the consistency of request handling. Automatic record creation, approval processing, status updates, and email notifications provide better visibility for both requesters and support teams.

Overall, the project demonstrates how ServiceNow can be used to build a **simple, efficient, and scalable network request management solution** that improves request tracking, reduces unnecessary follow-ups, and supports faster service delivery.

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THANK YOU!